

# Unsupervised Learning with discrete latent variable models

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# Organization

Tuesdays 9h45 - 13h (see agenda)

3 × 3h classes

**Come see me today for the reading report**

See course's website

# Bibliography & relevant sources

## General ML / Stats books

- Kevin P. Murphy (2022a). *Probabilistic Machine Learning: An introduction*. MIT Press
- Trevor Hastie et al. (2001). *The Elements of Statistical Learning*. Springer Series in Statistics. New York, NY, USA
- Christopher M. Bishop (2007). *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Springer

Some relevant lecture/slides on the topic for a different point-of-view ( $\triangle$  notations)

- S. Robin lectures
- Some lectures of this course on Graphical Models

# Introduction

# Types of statistical learning

## Supervised

Data  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$  with  $y_i$  an output (*response*) and  $x_i$  some features (*covariates*).

The goal is to learn a good predictor  $\hat{f}$  such that  $y_i \approx \hat{f}(x_i)$  that generalizes well on new data.

## Unsupervised (this course)

The data  $\mathcal{D} = \{x_i\}_{i=1}^n$  The goal is to learn "interesting" and hidden structure in the data to

- partition the data, aka clustering
- visualize/compress the data, aka dimension reduction

**Generative models:** posit a statistical model on the distribution of  $(X_i)$

## Many flavors in modern ML

semi-supervised, self-supervised, reinforcement learning, multi-task, etc.

# What this course is about...

Latent variables models for unsupervised learning

↪ we will assume the generative process of  $X$  involves an unobserved (latent) variable  $Z$

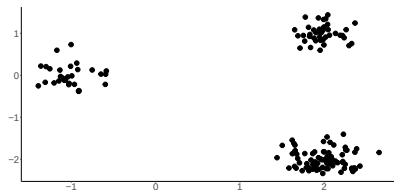
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$X$  is an unlabeled observation and  $Z$  its group membership



- K-means
- Mixture Models
- (hierarchical clustering, HMMs, graphs)

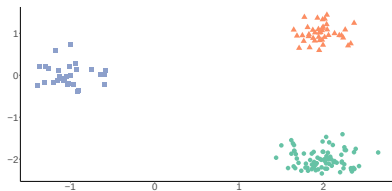
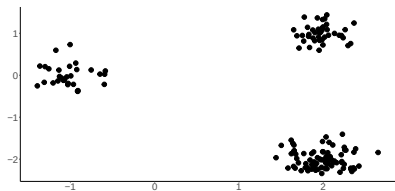
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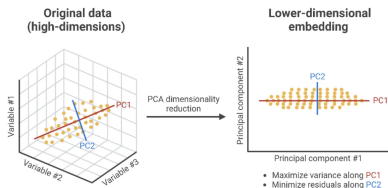
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## Example: Dimension reduction

$X$  in dimension  $p \gg 1$  and  $Z$  its low-dimensional representation



Source: <https://aiml.com/what-is-dimensionality-reduction-2/>

- Principal Component Analysis (PCA)
- Variational Auto-encoder (VAE)
- (UMAP, t-SNE)

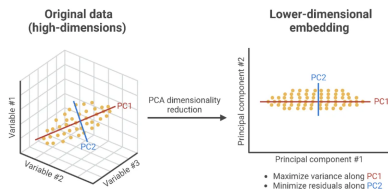
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# Incomplete data models

Most often, the observations are involved in complicated (biological, ecological, physical) processes, with many unobserved variables and complex dependency structure.

- $X$  observed random variables
- $Z$  unobserved (latent/hidden) variables
- $\theta$  unknown parameters

## An attempt at defining latent variables (creds. to S. Robin)

- Frequentist setting:

latent variables = random but unobserved, parameters = fixed

- Bayesian setting:

both latent variables and parameters = random

but

$\# \text{ latent variable} \simeq \# \text{ data}, \quad \# \text{ parameters} \ll \# \text{ data}$

# Different types of likelihoods

**In this course**, we place ourselves in the frequentist setting, using MLE inference. Although Bayesian extension of the proposed models are common.

## Complete data likelihood

Joint likelihood of the whole random process  $(\mathbf{X}, \mathbf{Z})$  with given parameters  $\theta$ .

$$p_{\theta}(\mathbf{X}, \mathbf{Z}) = p_{\theta}(\mathbf{X} \mid \mathbf{Z})p_{\theta}(\mathbf{Z}).$$

$\rightsquigarrow$  tractable in many models, but we do not observe  $\mathbf{Z}$  !

## Observed data likelihood

Marginal likelihood of the observed random variables  $\mathbf{X}$

$$p_{\theta}(\mathbf{X}) = \int_{\mathcal{Z}} p_{\theta}(\mathbf{X}, \mathbf{z}) d\mathbf{z}^a$$

$\rightsquigarrow$  only involves the observed  $\mathbf{X}$ , but not always tractable.

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<sup>a</sup>When  $\mathcal{Z}$  is discrete, replace  $\int$  by  $\sum$

# Course outline

- 1 Clustering with mixture models**
- 2 Inference in latent variable models: the EM algorithm**
- 3 Probabilistic dimension reduction**
- 4 Conclusion of the course**

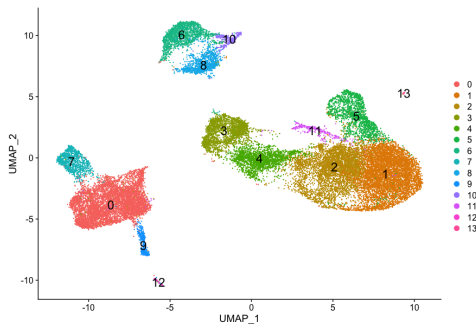
# Clustering with mixture models

# Motivation

Sometimes our data is organized in sub-population: groups of individuals we call *clusters*.

## Example

In modern biology, discovering cell-types via their gene expression profile is an important task.



When the groups are unknown, we call the task of discovering them *clustering*<sup>1</sup>

---

<sup>1</sup>as opposed to classification in a supervised context

# Mathematical context

We search for an optimal partition of  $\mathbf{x} = \{x_1, \dots, x_n\}$  into  $K$  groups.

## Definition: partition

A partition  $\mathcal{C} = \{C_1, \dots, C_K\}$  of  $\{1, \dots, n\}$  is a set of sets s.t.

$$\bigcup_k C_k = \{1, \dots, n\}, \quad \forall k \neq l, \quad C_k \cap C_l = \emptyset.$$

## Alternative encoding of the partition

For each individual  $i = 1, \dots, n$ , we define its *cluster membership*  $z_i \in \{0, 1\}^K$

$$k = 1, \dots, K, \quad z_{ik} = \begin{cases} 1 & \text{if } i \text{ belongs to cluster } k, \\ 0 & \text{otherwise} \end{cases}.$$

The set  $\mathbf{Z} = \{z_1, \dots, z_n\}$  represents a partition of  $\{1, \dots, n\}$ . This particular encoding is sometimes referred to as one-hot encoding.

# Clustering criteria

"Optimality" implies the definition of some criterion  $L \iff$  assumptions on the nature of clusters. Methods can be roughly split in two

## Similarity-based methods

Design  $L$  via geometric notions of similarity between  $x_i$ 's, favoring e.g.

- elliptic clusters
- convex clusters
- connected clusters

## Statistical methods

Consider the partition  $Z$  as a latent variable and posit a generative model  $p_\theta(\mathbf{X}, Z)$

$\rightsquigarrow$  Clustering becomes an inference problem of finding  $\hat{Z}$ .

There are connections between both !

# K-means

# The K-means problem

K-means seeks clusters well concentrated around their centroids  $\mu_k := \frac{1}{|C_k|} \sum_{i \in C_k} x_i$  by minimizing

$$\arg \min_C \left\{ L(C, \mathbf{X}) = \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|_2^2 \right\} \quad (\text{K-means problem})$$

- Good news: discrete problem  $\rightsquigarrow$  there exists an optimum  $C^\star$ .
- **Bad news:** there are  $K^n$  possible partitions  $\rightsquigarrow$  enumeration is not an option.

In fact, K-means problem is a **nonconvex NP-hard** problem and one need to resort to fast heuristics.

⚠ With a slight abuse, we drop distinction between K-means problem and heuristics to solve it.

# The K-means algorithm (MacQueen 1967)

Draw centroids  $\mu_1, \dots, \mu_K$  at random among the sample  $\mathbf{X}$  and

- 1 Assign each point to its closest centroid (Voronoi cells)

$$C_k \leftarrow \left\{ i : \|x_i - \mu_k\|_2^2 = \min_l \|x_i - \mu_l\|_2^2 \right\}$$

- 2 recompute centroids as the barycenter of each center

$$\mu_k := \frac{1}{|C_k|} \sum_{i \in C_k} y_i$$

- 3 Go to 1 until clusters (hence barycenters) are unchanged

## Properties of the algorithm

K-means is a greedy algorithm which

- monotonically decreases the criterion
- converges in a finite number of iterations
- will get stuck in local minima of  $L$  (non-convex)

↪ In practice, we try several restarts with different random inits.

# Extensions

**Kmeans++** initialization matter !  $\leadsto$  stop drawing centroids at random

- Choose  $\mu_1$  uniformly among the sample
- then sequentially do for each  $k = 2, \dots, K$ 
  - compute weight  $w_i := \min_{j < k} \|x_i - \mu_j\|_2^2$
  - Choose  $\mu_k$  among the sample with proba  $\propto w_i$

Optimality bounds can be obtained ([Arthur et al. 2007](#))

**Sparse K-means** include variable selection, useful when  $x_i$  in dimension  $d \gg n$

**Kernel K-means** compute distance between  $\phi(x_i)$  with  $\phi : \mathcal{X} \rightarrow \mathcal{H}$  a *feature map*.

# Mixture models

# Probabilistic view on clustering

The partition is now seen as a set of discrete latent variables  $\mathbf{Z} = \{z_1, \dots, z_n\}$

Denote  $\pi = (\pi_1, \dots, \pi_K)$  the (unknown) cluster proportions, we have

$$p_\pi(z_{ik} = 1) = \pi_k \iff z_i \sim \mathcal{M}(1, \pi)$$

## Mixture models

For all  $i = 1, \dots, n$ , mixture models suppose that  $(z_i, x_i)$  are drawn *i.i.d.* according to the two-stage hierarchical model

1  $Z_i \sim \mathcal{M}_K(1, \pi)$

2  $X_i \mid \{z_{ik} = 1\} \sim p_{\gamma_k}$

The model parameters are  $\theta = \{\pi_k, \gamma_k\}_{k=1}^K$  and  $p_\gamma$  can be any parametric distribution over  $X_i$ .

Clusters are sometimes called *components*

$\rightsquigarrow$  **general** and **flexible** framework, adapt to nature of the data (discrete, continuous, mixed-type) via  $p_\gamma$

# Observed (marginal) likelihood

## Properties: independence

In a mixture model,  $(Z_i)_i$  are *i.i.d.* and  $(X_i)_i$  also are *i.i.d.*

## Observed likelihood

$$\begin{aligned} p_{\theta}(\mathbf{X}) &= \sum_{z_1, \dots, z_n} p_{\theta}(\mathbf{Z}, \mathbf{X}) = \sum_{z_1, \dots, z_n} \prod_{i=1}^n p_{\theta}(X_i | z_i) p_{\theta}(z_i), \\ &= \prod_{i=1}^n \sum_{z_i} p_{\gamma}(X_i | z_i) p_{\theta}(z_i), \\ &= \prod_{i=1}^n \left( \sum_{k=1}^K \pi_k p_{\gamma_k}(X_i) \right). \end{aligned}$$

$\rightsquigarrow$  the marginal distribution of  $X_i$  is a convex combination (*mixture*) of the  $K$  base distributions  $(p_{\gamma_k})_k$ , with weights  $\pi_k$ .

# Complete likelihood

## Properties: conditional independence

In a mixture model,  $(X_i)_i \perp\!\!\!\perp \mathbf{Z}$  and  $(Z_i)_i \perp\!\!\!\perp \mathbf{X}$ , **but not** identically distributed

## Complete log-likelihood

$$\begin{aligned}\log p_{\theta}(\mathbf{X}, \mathbf{Z}) &= \log p_{\theta}(\mathbf{Z}) + \log p_{\theta}(\mathbf{X} \mid \mathbf{Z}) = \sum_{i=1}^n \log p_{\pi}(Z_i) + \log p_{\gamma}(X_i \mid Z_i), \\ &= \sum_{k=1}^K \sum_{i=1}^n Z_{ik} [\log \pi_k + \log p_{\gamma_k}(X_i)].\end{aligned}$$

## Posterior distribution of $\mathbf{Z} \mid \mathbf{X}$

For  $i = 1, \dots, n$ ,  $Z_i \mid X_i \sim \mathcal{M}_K(1, \tau_i)$  with

$$\tau_{ik} := p_{\theta}(z_{ik} = 1 \mid X_i) \propto \pi_k p_{\gamma_k}(X_i)$$

Notice that  $\tau_i$  also depends on the parameters  $\theta$ .

# A note on identifiability

## Definition: identifiability

A statistical model  $p_\theta$  is said to be identifiable iff the mapping  $\theta \mapsto p_\theta$  is injective.

**Intuition:** the labels of the clusters  $1, \dots, K$  should have no impact on the marginal likelihood

$$\pi_1 p_{\gamma_1}(x) + \pi_2 p_{\gamma_2}(x) = \pi_2 p_{\gamma_2}(x) + \pi_1 p_{\gamma_1}(x)$$

## Label switching

Let  $\sigma$  be a permutation of  $\llbracket 1, K \rrbracket$ , then for a mixture model with parameters  $\pi, \gamma$  we have

$$p(\mathbf{X} \mid \pi, \gamma) = p(\mathbf{X} \mid \sigma(\pi), \sigma(\gamma))$$

Hence, there are  $K!$  equivalent formulations of a mixture model.

↪ conceptually not a problem, it simply states that there are  $K!$  different encoding  $\mathbf{Z}$  of a given partition  $C = \{C_1, \dots, C_K\}$ .

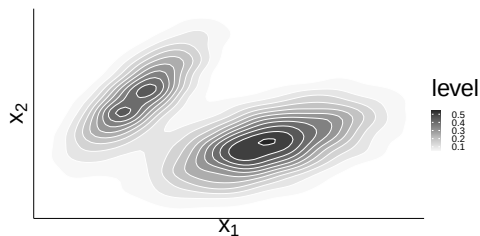
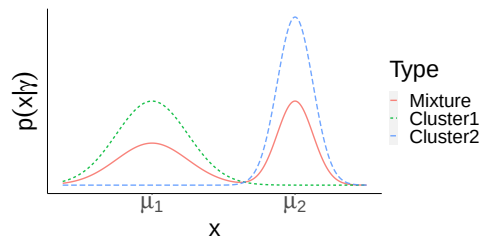
↪ can cause problems in Bayesian inference procedure since the posterior is highly multimodal.

# Gaussian Mixture Models (GMM)

**Continuous data:**  $x = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$

**Model:** Mixture of Gaussians  $p_{\gamma_k}(x) = \mathcal{N}(x \mid \mu_k, \Sigma_k)$ , with  $\gamma_k = (\mu_k, \Sigma_k)$

Multimodal marginal density around the  $(\mu_k)_k$ 's



**Number of free parameters:**  $K - 1 + Kd + K \frac{d(d+1)}{2} = \mathcal{O}(Kd^2)$  to estimate

# Maximum-likelihood estimation

Non-convex MLE problem

$$\arg \max_{\pi_k, \mu_k, \Sigma_k} \sum_{i=1}^n \log \left( \sum_{k=1}^K \pi_k \log \mathcal{N}(x_i \mid \mu_k, \Sigma_k) \right).$$

- Much more complex to maximize than in standard Gaussian models ( $K = 1$ )
- No closed-form solution, gradients can be derived but
  - 1 they are not cheap to compute at each iteration (although one could resort to stochastic optimization to leverage this issue).
  - 2 Requires re-projecting on the cone of p.d. matrices  $\Sigma_k \succ 0$ .

By contrast, the complete log-likelihood is much simpler to handle

$$\log p_{\theta}(\mathbf{x}, \mathbf{Z}) = \sum_{k=1}^K \sum_{i=1}^n Z_{ik} [\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \Sigma_k)].$$

~> **But we do not observe the  $\mathbf{Z}$  !**

# Maximum-likelihood estimation (cont'd)

## A chicken-and-egg problem

- 1 If we knew  $\mathbf{Z}$  we could maximize  $p_{\theta}(\mathbf{X}, \mathbf{Z}) \rightsquigarrow$  amount to compute MLE  $\hat{\gamma}_k$  in each cluster. In the Gaussian case we'd get cluster's empirical means and covariance

$$n_k = \sum_i z_{ik}, \quad \hat{\pi}_k = \frac{n_k}{n}, \quad \hat{\mu}_k = \sum_i z_{ik} x_i / n_k, \quad \hat{\Sigma}_k = \sum_i z_{ik} \frac{(x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^{\top}}{n_k}.$$

- 2 If we knew  $\theta^*$ , we could find the best estimate of  $\mathbf{Z}$  via the conditional distribution

$$\tau_{ik}(\theta) = p_{\theta}(z_{ik} = 1 \mid x_i) = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)}$$

$\rightsquigarrow$  this suggest an iterative scheme between 1) & 2) to solve MLE.

# Inference in latent variable models: the EM algorithm

## Some tools from information theory

# Jensen's inequality

**Quizz !** Which is larger:  $\mathbb{E}[Z^2]$  or  $\mathbb{E}[Z]^2$  ?

# Jensen's inequality

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**Answer:**  $\mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = \mathbb{V}(Z) \geq 0$

# Jensen's inequality

**Quiz !** Which is larger:  $\mathbb{E}[Z^2]$  or  $\mathbb{E}[Z]^2$  ?

**Answer:**  $\mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = \mathbb{V}(Z) \geq 0$

## General result: Jensen's inequality

Let  $Z$  be a random vector in  $\mathcal{Z} \subset \mathbb{R}^d$  and  $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$  a convex function, then

$$\mathbb{E}_Z [\phi(Z)] \geq \phi (\mathbb{E}_Z[Z]) . \quad (\text{Jensen})$$

$\rightsquigarrow$  the inequality is reversed with  $\phi$  concave ( $\phi \leftarrow -\phi$ )

*Proof:*

- $\phi$  convex  $\implies$  it is above its tangents, hence at any point  $z_0 \in \mathbb{R}^d, \exists a$  s.t.

$$\forall z \in \mathbb{R}^d, \quad \phi(z) \geq \phi(z_0) + a(z - z_0).$$

- Take  $z_0 = \mathbb{E}_Z[Z]$ , since the above inequality is true for all  $z$ , it generalizes to  $\mathbb{E}_Z$

$$\mathbb{E}_Z [\phi(Z)] \geq z_0 + \underbrace{a(\mathbb{E}_Z[Z] - z_0)}_{=0} = z_0 = \phi (\mathbb{E}_Z[Z])$$

# Entropy of a discrete random variable

## Definition: discrete entropy

For a discrete random variable  $Z$  with distribution  $q(Z = z)$  we define its entropy as

$$\mathcal{H}(Z) = \mathcal{H}(q) = -\mathbb{E} [\log q(Z)] = - \sum_{z \in \mathcal{Z}} q(z) \log q(z)$$

with the convention that  $0 \times \log 0 = 0$

## Properties

- $\mathcal{H}(q) \geq 0$
- **Continuous formulation:** Let  $Z$  be a r.v. with distribution  $Q$ . If there exist a measure  $\mu$  such that  $dQ = q d\mu$  then we can define

$$\mathcal{H}(Q) = \mathcal{H}_\mu(q) = - \int \log q(z) q(z) d\mu(z)$$

Now depends on the base measure  $\mu$ . Can be negative.

# Kullback-Leibler (KL) divergence

## Definition: KL divergence (discrete case)

Let  $p$  and  $q$  be two distribution over discrete set  $\mathcal{Z}$ , we define the KL-divergence as

$$\text{KL}(p \parallel q) := \mathbb{E}_{Z \sim p} \left[ \log \frac{p(Z)}{q(Z)} \right] = \sum_{z \in \mathcal{Z}} p(z) \log \frac{p(z)}{q(z)}$$

## Properties

- $\text{KL}(p \parallel q) \geq 0$  with equality iff  $p = q$  (*proof*: Jensen on  $\frac{p}{q}(Z)$  with convex  $\phi(x) = -\log x$ )
- Diverges if  $\exists z_0$  such that  $q(z_0) = 0$  when  $p(z_0) \neq 0$
- Not a distance (not symmetric)
- **Continuous formulation:** For two distribution  $P$  and  $Q$ , if there exists a measure  $\mu$  such that  $dP = p d\mu$  and  $dQ = q d\mu$ , then

$$\text{KL}(P \parallel Q) = \int \log \frac{dP}{dQ} dP = \int \log \frac{p(z)}{q(z)} p(z) d\mu(z).$$

$\rightsquigarrow$  invariant w.r.t. the choice of  $(p, q, \mu)$  since the ratio  $dP/dQ$  is invariant.

## The evidence lower bound (ELBO)

# Minorizer of the observed-likelihood

## Evidence lower bound

Let  $q$  be a distribution over  $\mathcal{Z}$  absolutely continuous with respect to  $p_\theta(X, Z)$ . Then,

$$\log p_\theta(\mathbf{X}) \geq \mathcal{L}(q, \theta) := \mathbb{E}_q [\log p_\theta(X, Z)] + \mathcal{H}(q). \quad (\text{ELBO})$$

The quantity  $\mathcal{L}$  is called *the evidence lower-bound*, moreover the gap is expressed as

$$\log p_\theta(X) - \mathcal{L}(q, \theta) = \text{KL}(q \parallel p_\theta(\cdot \mid X)).$$

$$\text{Proof: } \log p_\theta(X) = \log \int p_\theta(X, z) \, dz = \log \mathbb{E}_q \left[ \frac{p_\theta(X, Z)}{q(Z)} \right] \stackrel{\text{Jensen}}{\geq} \mathbb{E}_q \left[ \log \frac{p_\theta(X, Z)}{q(Z)} \right] = \mathcal{L}(q, \theta)$$

## Comments

- The ELBO holds for any distribution  $q$  on  $\mathcal{Z}$
- For a given  $\theta$ , the gap is 0 iff

$$q(z) = p_\theta(z \mid X)$$

## Expectation-maximization (EM, Dempster et al. 1977)

# EM: a universal algorithm for latent variables

**Intuition:** chicken-and-egg

- 1 if we knew  $\mathbf{Z}$ , we could easily work with  $f(\theta) = \log p_{\theta}(\mathbf{X}, \mathbf{Z})$
- 2 if we knew  $\theta$ , the best representation of  $\mathbf{Z}$  is via the conditional  $p_{\theta}(\mathbf{Z} \mid \mathbf{X})$

## Expectation-Maximization algorithm

Starting from  $\theta^{(0)}$ , iterate between

### Expectation step

Use  $q^{(t+1)}(\mathbf{Z}) = p_{\theta^{(t)}}(\mathbf{Z} \mid \mathbf{X})$  to form the objective function

$$f(\theta) = Q(\theta, \theta^{(t)}) = \mathbb{E}_{\mathbf{Z} \sim q^{(t+1)}} [\log p_{\theta}(\mathbf{X}, \mathbf{Z})] .$$

It involves (generalized) moments of  $\mathbf{Z}$  under  $q^{(t+1)}$ .

### Maximization step

Solve  $\theta^{(t+1)} \in \arg \max_{\theta} Q(\theta, \theta^{(t)})$

In practice, EM stop after likelihood gaps fall below a given threshold  $\epsilon$

$$|\mathcal{L}(q^{(t+1)}, \theta^{(t)}) - \mathcal{L}(q^{(t)}, \theta^{(t-1)})| = |\log p_{\theta^{(t)}}(\mathbf{X}) - \log p_{\theta^{(t-1)}}(\mathbf{X})| < \epsilon$$

# Monotonic increase of the observed likelihood

## Property of EM algorithm

The sequence of iterates  $\{\theta^{(t)}\}_t$  returned by EM verifies

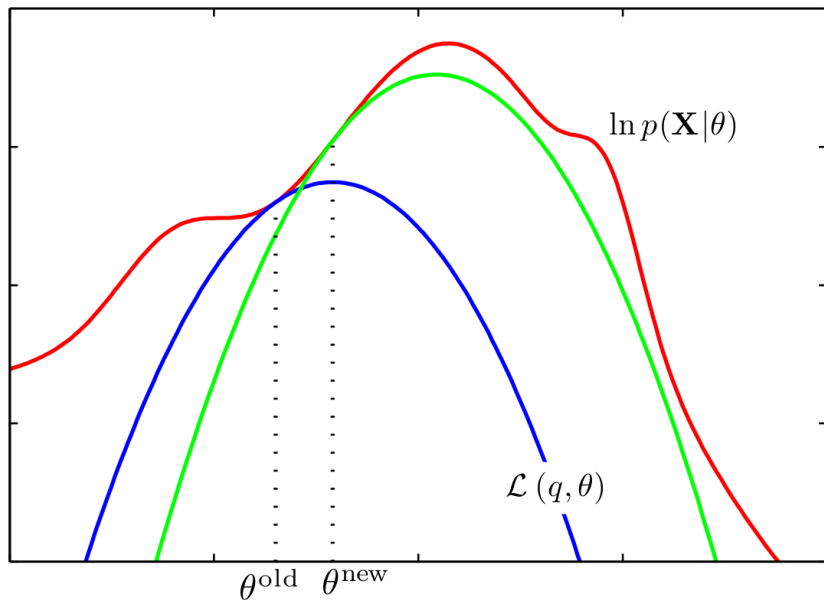
$$\forall t \geq 0, \quad \log p_{\theta^{(t+1)}}(\mathbf{X}) \geq \log p_{\theta^{(t)}}(\mathbf{X})$$

*Proof:*

$$\log p_{\theta^{(t+1)}}(\mathbf{X}) \underbrace{\geq}_{\text{ELBO}} \mathcal{L}(q^{(t+1)}, \theta^{(t+1)}) \underbrace{\geq}_{\text{M-step}(t+1)} \mathcal{L}(q^{(t+1)}, \theta^{(t)}) \underbrace{=}_{\text{E-step}(t)} \log p_{\theta^{(t)}}(\mathbf{X})$$

- Guarantees EM converges with the likelihood gaps criterion
- In general, only converges to local maxima of the likelihood
- Does not guarantee convergence of the sequence of parameters  $\{\theta^{(t)}\}_t$  itself.

# A graphical illustration of EM algorithm (cred: G. Obozinski)



Source: [https://www.iro.umontreal.ca/~slacoste/teaching/MVA\\_GM/fall2015/cours\\_MVA\\_EM.pdf](https://www.iro.umontreal.ca/~slacoste/teaching/MVA_GM/fall2015/cours_MVA_EM.pdf)

## Illustration with Gaussian mixture

# Expected complete log-likelihood

Denote  $\tau_{ik}^{(t)} := p_{\theta^{(t-1)}}(Z_{ik} = 1 \mid x_i) \underset{\text{Multinomial}}{=} \mathbb{E}_{q^{(t)}}[Z_{ik}]$ , then

$$\begin{aligned} f(\theta) &= \mathbb{E}_{q^{(t)}} [\log p_{\theta}(\mathbf{X}, \mathbf{Z})] , \\ &= \mathbb{E}_{q^{(t)}} \left[ \sum_{i=1}^n \log p_{\theta}(x_i, Z_i) \right] , \\ &= \mathbb{E}_{q^{(t)}} \left[ \sum_{k=1}^K \sum_{i=1}^n Z_{ik} [\log \pi_k + \log \mathcal{N}_q(x_i \mid \mu_k, \Sigma_k)] \right] , \\ &= \sum_{k=1}^K \sum_{i=1}^n \mathbb{E}_{q_i^{(t)}} [Z_{ik}] [\log \pi_k + \log \mathcal{N}_d(x_i \mid \mu_k, \Sigma_k)] , \\ &= \sum_{k=1}^K \sum_{i=1}^n \tau_{ik}^{(t)} [\log \pi_k + \log \mathcal{N}_d(x_i \mid \mu_k, \Sigma_k)] , \end{aligned}$$

It involves  $\tau_{ik}^{(t)}$  : (first) moments of  $Z$  under  $q^{(t)}$ .

# E-step for GMM

**Compute the conditional** given  $\theta^{(t-1)}$ ,  $q^{(t)} = p_{\theta^{(t-1)}}(\mathbf{Z} \mid \mathbf{X})$

As seen previously, the conditional for mixture model always writes

$$p_{\theta}(\mathbf{Z} \mid \mathbf{X}) = \prod_{i=1}^n \mathcal{M}_K(1, \tau_i(\theta)), \quad \text{with: } \tau_{ik}(\theta) \propto \pi_k p_{\gamma_k}(x_i).$$

So that

$$\tau_{ik}^{(t)} = \tau_{ik}(\theta^{(t-1)}) = \frac{\pi_k \mathcal{N}_d(x_i \mid \mu_k^{(t-1)}, \Sigma_k^{(t-1)})}{\sum_{l=1}^K \pi_l \mathcal{N}_d(x_i \mid \mu_l^{(t-1)}, \Sigma_l^{(t-1)})}.$$

**Careful** with numerical underflow  $\rightsquigarrow$  better to work with in log-space with  $\log \tau$ .

# M-step for GMM

Solve

$$(\pi_k^{(t)}, \mu_k^{(t)}, \Sigma_k^{(t)})_{k=1}^K \in \arg \max_{\substack{\theta = (\pi_k, \mu_k, \Sigma_k)_k \\ \sum_k \pi_k = 1}} \left\{ f(\theta) = \mathbb{E}_{q^{(t)}} [\log p_{\theta}(\mathbf{X}, \mathbf{Z})] \right\}$$

For GMM, using the KKT conditions accounting for constraints on  $\pi$  the updates are

$$\begin{cases} \tilde{n}_k^{(t)} = \sum_{i=1}^n \tau_{ik}^{(t)}, \\ \pi_k^{(t)} = \frac{\tilde{n}_k^{(t)}}{n}, \\ \mu_k^{(t)} = \frac{1}{\tilde{n}_k^{(t)}} \sum_{i=1}^n \tau_{ik}^{(t)} x_i, \\ \Sigma_k = \frac{1}{\tilde{n}_k^{(t)}} \sum_{i=1}^n \tau_{ik}^{(t)} (x_i - \mu_k^{(t)})(x_i - \mu_k^{(t)})^{\top} \end{cases}$$

We recognize standard Gaussian MLE in each cluster, using soft probability memberships  $\tau$  in place of unknown  $\mathbf{Z}$ .

## Link with K-means algorithm

The K-means algorithm can be interpreted as an EM algorithm for a constrained GMM with equal proportions  $\pi_k = 1/K$ , known isotropic covariance  $\Sigma_k = \sigma^2 \text{Id}_d$ . Dropping the known quantities, the criterion is

$$\arg \min_{\mu_1, \dots, \mu_K, \mathbf{Z}} -\log p_{\mu}(\mathbf{X}, \mathbf{Z}) = cte + \sum_k \sum_{i \in C_k} \|x_i - \mu_k\|_2^2.$$

### Rewriting K-means (Classification-EM for GMM)

- 1 *Hard E-step*: set partition  $C^{(t+1)}$  via MAP  $\arg \max_l \tau_{il}^{(t+1)} = \arg \min_l \|x_i - \mu_l^{(t)}\|_2^2$
- 2 *M-step*: update the centroids  $\mu_k^{(t+1)} \leftarrow (1/n_k) \sum_{i \in C_k^{(t+1)}} x_i$

### Comments

- highlight connections between similarity-based and probabilistic methods
- unveil hypothesis behind K-means criterion: spherical, equal-volume and equal-size clusters.

# Choosing the number of components $K$

**Challenge:** how to choose the number of clusters  $K$  ?

**Intuition:** the larger the likelihood, the better our model fits the data  $\mathbf{X}$

**Caveat:** complex models tend to provide larger likelihood, for example

- mixture models with  $K - 1$  components are nested in models with  $K$  components.
- models with constraints (diagonal, spherical) are nested in unconstrained ones.

⇒ we need to account for "model complexity"

**Definition: dimension/size of a model**

Let  $\mathcal{M} = \{p_\theta, \theta \in \Theta_{\mathcal{M}}\}$ , we denote  $d_{\mathcal{M}}$  the number of free parameters in the model. For unconstrained mixtures, it is  $d_K = K - 1 + Kd_\Gamma$ ,  $\gamma_k \in \Gamma$ .

## Penalized likelihood criterion

For a mixture model with  $K$  components, denote  $\hat{\theta}_K = \arg \max_{\theta \in \Theta_K} \log p_\theta(\mathbf{X})$ . A *penalized likelihood* estimate of  $K$  is given by

$$\hat{K} = \arg \max_K \left\{ \log p_{\hat{\theta}_K} - \text{pen}(K) \right\}.$$

# Different penalties leads to different criterion

## Definitions: AIC, BIC, ICL

For a model  $\mathcal{M}$  and observations  $\mathbf{X}$ , we have several choice of penalize likelihood criteria

$$AIC(K) := \log p_{\hat{\theta}_K}(\mathbf{X}) - d_K,$$

$$BIC(K) := \log p_{\hat{\theta}_K}(\mathbf{X}) - \frac{d_K}{2} \log(n),$$

$$ICL(K) := \mathbb{E}_{\mathbf{Z} \sim p_{\hat{\theta}_K}(\cdot | \mathbf{X})} \left[ \log p_{\hat{\theta}_K}(\mathbf{X}, \mathbf{Z}) \right] - \frac{d_K}{2} \log(n)$$

**Note:** the ELBO property gives

$$ICL(K) = BIC(K) - \mathcal{H}(p_{\hat{\theta}_K}(\cdot | \mathbf{X})).$$

Hence, ICL is more focused on models with strongly separable clusters (peaked conditional  $\Rightarrow$  low entropy), while BIC is more focused on fitting the marginal density of  $\mathbf{X}$ .

## Focus on BIC: Bayesian information criterion

Put a prior  $p(K)$  on  $K$ , and the model:  $p(\theta | K)$  and  $p(\mathbf{X} | \theta)$ . Bayes rule suggests choosing

$$\begin{aligned}\hat{K} &= \arg \max_K \{p(K | \mathbf{X}) \propto p(K)p(\mathbf{X} | \theta)\}, \\ &= \arg \max_K \log p(K) + \log p(\mathbf{X} | K), \\ &= \arg \max_K \log p(K) + \log \int p(\mathbf{X} | \theta, K)p(\theta | K) d\theta.\end{aligned}$$

Dropping the prior term  $\log p(K)$  which is constant with  $n$ , we need to compute the integral in the second term  $\rightsquigarrow$  difficult in general !

Under regularity assumptions (see [Lebarbier et al. 2004, for details](#)), we have

$$\log p(\mathbf{X} | K) = \log p_{\hat{\theta}_K}(\mathbf{X}) - \frac{d_K}{2} \log(n) + \mathcal{O}_P(1).$$

This justifies the formula of BIC.

# Probabilistic dimension reduction

# Principal component analysis (PCA)

# PCA as a distance-preserving projection

Observe  $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$  **centered**:  $\bar{x} = \mathbf{0}_d$

**Goal:** we seek

1 a low-dimensional representation  $z_i \in \mathbb{R}^q$

2 a linear transform  $W \in \mathbb{R}^{d \times q}$

with  $q \ll d$  and such that

$$x_i \approx W z_i, \quad \text{with: } W^\top W = Id_q$$

**How ?** Minimize euclidean distance, define  $\|A\|_F^2 = \text{Tr}(A^\top A)$

$$\hat{W}, \hat{Z} = \arg \min_{\substack{W^\top W = Id_q \\ Z \in \mathbb{R}^{n \times q}}} \left\{ L(W, Z) := \sum_{i=1}^n \|x_i - W z_i\|_2^2 = \|X - ZW^\top\|_F^2 \right\} \quad (\text{PCA})$$

**Note:** For any fixed  $W$ , minimizing convex  $z_i \mapsto \|x_i - W z_i\|_2^2$  leads to  $\hat{z}_i = W^\top x_i$ .

$$\implies \hat{Z} = XW$$

# PCA solution: SVD on $X$

## Eckart-Young-Mirsky theorem for best rank- $q$ approximation wrt $\|\cdot\|_F$

Let the svd of  $X = UDV^\top$  with sorted singular values  $D = \text{diag}(\sigma_1, \dots, \sigma_d)$ . Note the top- $q$  singular values  $D_q := \text{diag}(\sigma_1, \dots, \sigma_q)$ , and associated right (resp. left) singular vectors  $V_q$  (resp.  $U_q$ ). Then the matrix  $X_q = U_q D_q V_q^\top$  is solution of

$$\|X - X_q\|_F^2 = \min_{A \text{ s.t. } \text{rank}(A) \leq q} \|X - A\|_F^2$$

**Corollary** The solution of PCA is given by  $\hat{W} = V_q$  and  $\hat{Z} = X\hat{W} = XV_q = U_q D_q$

*Proof:* we have  $\text{rank}(ZW) \leq \text{rank}(Z) \leq q$  and  $\hat{W} = U_q$  reaches the bound

**Note:** right singular-vectors  $V \iff$  eigenvectors of  $\hat{\Sigma} = X^\top X$  the covariance matrix  
 $\rightsquigarrow$  coherent with the alternative formulation of variance maximizing projection!

## PCA-vocabulary:

- **Principal components:** the columns of  $W$  that span  $\mathcal{S} = \text{span}(W)$
- **Principal components scores:**  $z_i = \text{proj}_{\mathcal{S}}(x_i)$  are coordinates in the subspace.

## Probabilistic PCA (pPCA)

# A (noisy) low-rank Gaussian model

## Probabilistic PCA (pPCA)

Tipping et al. (1999b) proposed a latent variable formulation. For all  $i = 1, \dots, n$

$$z_i \sim \mathcal{N}_q(0_q, I_q), \quad (\text{Latent})$$

$$x_i = \mu + W z_i + \epsilon_i, \quad \text{with } \epsilon_i \sim \mathcal{N}_d(0_d, \sigma^2 I_d). \quad (\text{Observed})$$

With  $q \leq d$  and  $\theta = (\mu, W, \sigma^2) \in \mathbb{R}^d \times \mathbb{R}^{p \times d} \times \mathbb{R}_+$ .

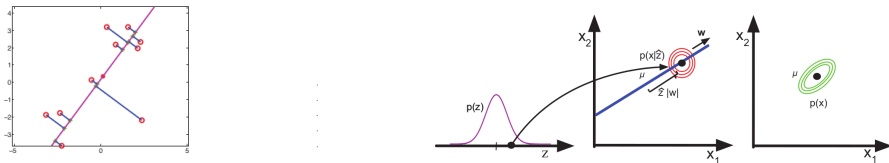
**Marginal distribution:** low-rank Gaussian

$$x_i \sim \mathcal{N}_d(\mu, C)$$

with marginal covariance  $C = WW^\top + \sigma^2 I_d$

**Number of free parameters:**  $d + dq + 1$

# A generative model of PCA



**Figure:** Figures from [Murphy \(2022b, Chap. 12\)](#): illustrations for  $d = 2$  and  $q = 1$ . PCA (left) projects observations to latent space, and probabilistic PCA (right) models the reverse process mapping Gaussians from latent to observed.

## A note on identifiability

For any rotation on the latent space  $RR^\top = R^\top R = I_q$ ,  $R \in \mathbb{R}^{q \times q}$ , the matrix  $\tilde{W} = WR$  yields the same marginal covariance

$$\tilde{C} = \tilde{W}\tilde{W}^\top + \sigma^2 I_d = WRR^\top W^\top + \sigma^2 I_d = C$$

Thus, the matrix  $W$  of pPCA is only identifiable up to a rotation of the principal components.

# MLE for pPCA

## Observed-likelihood

$$\log p_{\theta}(X) = -\frac{n}{2} \log |C| - \frac{1}{2} \sum_{i=1}^n (x_i - \mu) C^{-1} (x_i - \mu) = -\frac{n}{2} \left[ \log |C| + \text{Tr}(C^{-1} \hat{\Sigma}) \right],$$

with  $\hat{\Sigma} := n^{-1} \sum_i (x_i - \mu)(x_i - \mu)^{\top}$  is the empirical covariance.

## Maximum-likelihood (Tipping et al. 1999a, Sec 3.2)

$$(\hat{\mu}, \hat{W}, \hat{\sigma}^2) = \arg \max_{\mu, W, \sigma^2 > 0} \log p_{(\mu, W, \sigma^2)}(X)$$

is such that, for any rotation  $R^{\top} R = I_q$

$$\hat{\mu} = \bar{\mu} := \frac{1}{n} \sum_{i=1}^n x_i \quad (\text{Emp. mean})$$

$$\hat{W} = V_q (D_q^2 - \sigma^2 I_q)^{1/2} R \quad (\text{PC} \times \text{eigvals})$$

$$\hat{\sigma}^2 = \frac{1}{d - q} \sum_{j=q+1}^d \sigma_j^2 \quad (\text{avg} \perp \text{variance})$$

**Note:** when  $\sigma^2 \rightarrow 0$  we recover deterministic PCA !

## EM algorithm for pPCA

# Why an EM algorithm ?

We already have the MLE for  $\hat{W}$ . Why bother with an EM ?

- 1 If  $d$  or  $n$  are big  $\rightsquigarrow$  the updates do not require storing  $\hat{\Sigma}$
- 2 EM can handle missing data  $\rightsquigarrow$  common with real datasets
- 3 Extensions of pPCA do not have explicit MLE
  - Mixture of pPCA
  - "Factor Analysis": non-isotropic noise  $\text{Cov}(\epsilon_i) = \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$
  - Non-Gaussian prior on  $z_i \sim p_\theta(\cdot)$
  - Variational auto-encoders: non-linear mapping  $x_i = f_\theta(z_i) + \epsilon_i$

# M-step for pPCA

**E-step boils down to** given  $\theta^{(t-1)} = (\tilde{W}, \tilde{\sigma}^2)$  compute the conditional means  $m_i$  and

$$M := (\mathbf{X} - \mathbf{1}_n \bar{\mu}^\top) \tilde{W} \tilde{F}^{-\top}$$

**M-step** Maximize the objective

$$(W^{(t)}, \sigma^{2(t)}) = \arg \max_{\theta = (W, \sigma^2)} f(\theta) = \mathbb{E}_{p_{\theta^{(t-1)}}(\mathbf{Z}|\mathbf{X})} [\log p_{\theta}(\mathbf{X}, \mathbf{Z}) \mid \mathbf{X}]$$

which yield updates

$$W^{(t)} = \left\{ \sum_{i=1}^n (x_i - \bar{\mu}) m_i^\top \right\} \left\{ \sum_{i=1}^n \tilde{\sigma}^2 \tilde{F}^{-1} + m_i m_i^\top \right\}^{-1}, \quad (1)$$

$$\sigma^{2(t)} = \frac{1}{nd} \sum_{i=1}^n \left\{ \|x_i - \bar{\mu}\|_2^2 - 2 m_i^\top W^{(t)} (x_i - \bar{\mu}) + \text{Tr}[(\tilde{\sigma}^2 \tilde{F}^{-1} + m_i m_i^\top) W^{(t)\top} W^{(t)}] \right\}. \quad (2)$$

# Some notes on this EM algorithm

**Convergence** [Tipping et al. \(1999a\)](#) showed that the only stable stationary point of the likelihood is the MLE. The rest are saddle points

**Noiseless regime**  $\sigma^2 \rightarrow 0$  the algorithm remains valid ([Roweis 1997](#))

- **E-step:**  $m_i = (WW^\top)^{-1} W(x_i - \bar{\mu}) \rightsquigarrow \perp$  projection on  $\text{span}(W)$
- **M-step** Update for  $W^{(t)}$  are akin linear regression of  $\mathbb{E}[z_i|x_i]$  on  $x_i$ .

This is an iterative algorithm for deterministic PCA !

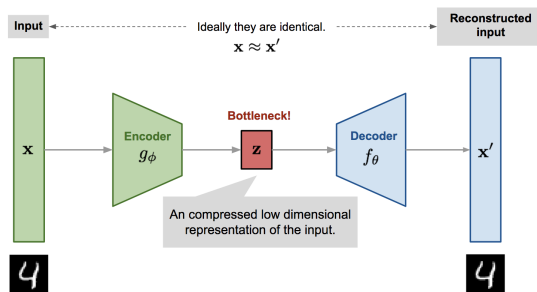
**Computational complexity** for  $T$  iterations of EM

- **E-steps** complexity of  $\mathcal{O}(Tndq) \rightsquigarrow$  to compare to the MLE which requires eigen decomposition of the sample covariance  $\hat{\Sigma}$  in  $\mathcal{O}(\min(nd^2, n^2d))$ . May be favorable if  $q \ll d$  and few iterations.

In practice, few people use EM for pPCA **BUT** useful to know for non-linear extensions.

## Non-linear extension: variational auto-encoders (VAEs)

# Deterministic DR: PCA and auto-encoders



$$\min_{\phi, \theta} \mathcal{L}(\phi, \theta) = \sum_{i=1}^n \|x_i - f_\theta(g_\phi(x_i))\|^2$$

## Auto-encoders

Two neural-net  $f_\theta, g_\phi$

$$z_i = g_\phi(x_i) \quad (\text{Encoding})$$

$$\hat{x}_i = f_\theta(z_i) \approx x_i \quad (\text{Decoding})$$

## PCA

Linear nets:  $\theta = W$  and  $\phi = W^\top$

$$z_i = g(x_i) = W^\top x_i$$

$$\hat{x}_i = f(z_i) = W z_i \approx x_i$$

# Probabilistic extension: variational AEs

**pPCA** probabilistic decoding & encoding  $z \sim \mathcal{N}_q(0, I_q)$

- **Decoder:** probabilistic linear mapping for mean  $\mu_W(z)$  of  $x|z$

$$x \mid z \sim \mathcal{N}_d(\mu_W(z), \sigma^2 I_d), \quad \text{with: } \mu_W(z) = Wz$$

- **Encoder:** tractable conditional mean  $\mu_\phi(x)$  is linear in  $x$

$$z \mid x \sim \mathcal{N}_q(\mu_\phi(x), \sigma^2 F^{-1}), \quad \text{with: } \mu_\phi(x) = \underbrace{F^{-1} W^\top}_{\phi} x$$

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**Variational auto-encoders ?** Use a non-linear mapping  $\mu_W \rightarrow f_\theta$  for the parameters of  $x \mid z$   
The emission law  $x \mid z$  can be anything, e.g.

- *Gaussian*  $\approx$  non-linear pPCA !

$$f_\theta : z \in \mathbb{R}^q \mapsto (\mu_\theta(z), \Sigma_\theta(z)) \in \mathbb{R}^d \times \mathcal{S}_+^d$$
$$x_i \mid z_i \sim p(x_i \mid f_\theta(z_i)) = \mathcal{N}(x_i \mid \mu_\theta(z_i), \Sigma_\theta(z_i))$$

- *Poisson*: count data  $x_i \in \mathcal{X} = \mathbb{N}$

$$x_i \mid z_i \sim p(x_i \mid f_\theta(z_i)) = \mathcal{P}(x \mid \lambda_\theta(z)), \quad \text{with: } \lambda_\theta(z) \in \mathbb{R}_+$$

**But what about  $p_\theta(z \mid x)$  ?**

# Intractable conditional $z \mid x$ in VAEs

**Bayes rule** we know that

$$p_{\theta}(z \mid x) = \frac{p_{\theta}(x \mid z)p(z)}{p_{\theta}(x)}$$

The normalization constant is generally intractable

$$p_{\theta}(x) = \int p(x \mid f_{\theta}(z))p(z) \mathrm{d}z$$

**Idea** approximate  $p_{\theta}(z \mid x)$  by another simpler distribution  $q_{\phi}(z \mid x)$  parametrized by a neural net  $g_{\phi}$ . Usually we take a Gaussian

$$q_{\phi}(z \mid x) = q(z \mid g_{\phi}(x)) = \mathcal{N}_q(z \mid \mu_{\phi}(x), s_{\phi}(x))$$

**But how to find the best  $q_{\phi}$  ?**

$$\min_{\phi} \mathrm{KL}(q(z \mid g_{\phi}(x)) \parallel p_{\theta}(z \mid x))$$

**Reminds you of something ?**  $\rightsquigarrow$  your favourite EM algorithm, yeaah!

# VAE as two neural net

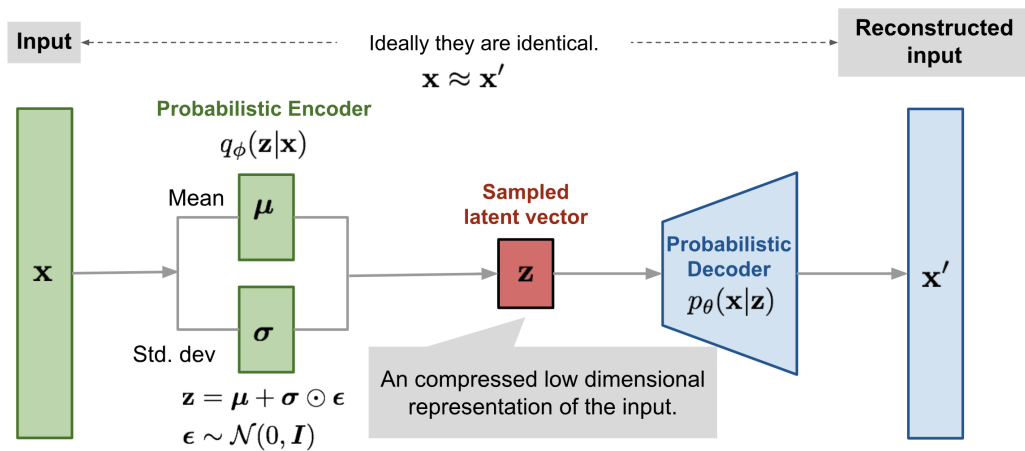


Image taken from [lilianweng.github.io](https://github.com/lilianweng)

# Summary

Summary of link between PCA and auto-encoder

| Method  | Projection type | Decoding                | Encoding        |
|---------|-----------------|-------------------------|-----------------|
| PCA     | Linear          | $\hat{x} = Wz$          | $z = W^\top x$  |
| Encoder | Non-linear      | $\hat{x} = f_\theta(z)$ | $z = g_\phi(x)$ |

Summary of link between pPCA and VAEs

| Method | Projection type | $x   z$ (Decoding)                               | $p(z   x)$ (Encoder)                                                 |
|--------|-----------------|--------------------------------------------------|----------------------------------------------------------------------|
| pPCA   | Linear          | $\mathcal{N}_d(Wz, \sigma^2 I)$                  | $\mathcal{N}_q(C^{-1} Wx, \sigma^2 C)$                               |
| VAEs   | Non-linear      | $\mathcal{N}_d(\mu_\theta(z), \Sigma_\theta(z))$ | $\approx_{KL} q_{g_\phi(x)} = \mathcal{N}_q(\mu_\phi(x), s_\phi(x))$ |

## Variational inference & illustration for the VAEs

## Untractable E-step

**Until now** we always managed to compute the necessary moments of the conditional for E-step, *i.e.* to compute (given  $\theta^{(t)}$ )

$$f(\theta) = \mathbb{E}_{\mathbf{Z} \sim p_{\theta^{(t)}}(\cdot | \mathbf{X})} [\log p_{\theta}(\mathbf{X}, \mathbf{Z})] \quad (3)$$

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**Reminders:** computing Equation (3) involve

- *For mixtures:* the marginal  $\tau_{ik}^{(t+1)} = p_{\theta^{(t)}}(z_i = k | \mathbf{X}) = p_{\theta^{(t)}}(z_i = k | x_i)$  (conditional independence).
- *For pPCA:* Gaussian world, the conditional is also Gaussian

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**Problem:** what if there's no hope of reasonable computation time for Equation (3) ? Either because

- 1 *complicated conditional dependencies:*  $z_1, \dots, z_n | \mathbf{X}$  (DAG moralization)
- 2 *intractable normalization constant*  $p_{\theta}(x_i) = \int p_{\theta}(x_i | \mathbf{Z}) d\mathbf{Z}$

# Variational inference: constraining the E-step

Since complex models have intractable conditionals, we need to constrain the distribution  $q$  to belong in some prescribed family of probability distributions<sup>2</sup>  $q \in \mathcal{Q} \subset \mathcal{P}(\mathbf{Z})$ .

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The *variational, E-step* becomes

$$q^{(t+1)} = \arg \max_{q \in \mathcal{Q}} \mathcal{L}(q, \theta^{(t)}). \quad (\text{VE-step})$$

Or, equivalently,

$$q^{(t+1)} = \arg \min_{q \in \mathcal{Q}} \text{KL}(q(\cdot) \parallel p_{\theta^{(t)}}(\cdot \mid \mathbf{X})).$$

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**Key idea:** choose  $\mathcal{Q}$  such that calculations in (VE-step) are tractable.

---

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# A common choice of variational family: mean-field approximation

**Natural follow-up question:** what choice for  $q$  ?

*Mean-field* family: "forget" conditional dependencies of  $\mathbf{Z} \mid \mathbf{X}$

$$q \in \mathcal{Q} = \left\{ q_{\tau} : q_{\tau}(\mathbf{Z}) = \prod_{i=1}^n q_{\tau_i}(z_i), \quad \tau_i \in \Psi \right\} \quad \text{so that} \quad \max_{q \in \mathcal{Q}} \mathcal{L}(q) = \max_{\tau \in \Psi^n} \mathcal{L}(q_{\tau}). \quad (4)$$

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**Important remark:**  $q$  is not a *model* of the observed data but rather the ELBO (and the KL minimization) connects  $q$  to the data & the model ([Blei et al. 2017](#))

# A common choice of variational family: mean-field approximation

**Natural follow-up question:** what choice for  $q$  ?

*Mean-field* family: "forget" conditional dependencies of  $\mathbf{Z} \mid \mathbf{X}$

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## Property

- Entropy term: by independence  $\mathcal{H}(q) = \sum_{i=1}^n \mathcal{H}(q_i)$
- we can enforce a parametric form
  - 1  $z_i$  is discrete (mixture):  $q_{\tau_i}(z_i) = \mathcal{M}_K(1; \tau_i)$  and  $\Psi = \Delta_K$
  - 2  $z_i$  continuous (pPCA, VAE):  $q_{\tau_i}(z_i) = \mathcal{N}_q(\mu_i, S_i)$  and  $\Psi = \mathbb{R}^q \times \mathcal{S}_+^q$

**Note:** what drives the choice of  $\mathcal{Q}$  is the practical calculations of  $\mathbb{E}_q[\cdot]$  under  $q \in \mathcal{Q} \rightsquigarrow$  model dependent

# Variational-EM (VEM) algorithm

## VEM algo: coordinate-ascent on the ELBO

Start from initial  $\theta^{(0)}$  and set a variational family  $\mathcal{Q}$

$$q^{(t+1)} = \arg \max_{q \in \mathcal{Q}} \mathcal{L}(q, \theta^{(t)}), \quad (\text{VE-step})$$

$$\theta^{(t+1)} = \arg \max_{\theta} \mathcal{L}(q^{(t+1)}, \theta), \quad (\text{M-step})$$

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$$q_i^* = \arg \max_{q_i} \mathcal{L}(q_i, q_{-i}). \quad (\text{CAVI})$$

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Pros & cons of VEM algorithm

### ■ Pros:

- 1 we choose  $\mathcal{Q}$  such that everything is tractable
- 2 Approximation of intractable conditional via  $q^{(T)}$  in the sense of KL-divergence

■ Cons: only increase the ELBO, no guarantee to increase the likelihood anymore ! We get an estimator

$$\hat{\theta}_V \in \arg \max_{\theta} \mathcal{L}(q^{(T)}, \theta)$$

# Conclusion of the course

# What we saw in this course

**Three examples of general discrete latent variable models:** GMM, pPCA, VAEs

- incomplete data models:  $p_{\theta}(\mathbf{X}) = \sum_{\mathbf{Z}} p_{\theta}(\mathbf{X}, \mathbf{Z})$
- the complete likelihood is easier to write than the marginal (but we do not observe  $\mathbf{Z}$ )
- Generalizes well to different type of data (discrete, continuous) via the choice of different observation model  $\mathbf{X} | \mathbf{Z}$  (i.e.  $p_{\gamma}$ )

**Link between machine learning methods and statistical model** (explicit assumptions about data)

**Inference procedures** for latent variable model

- EM algorithm
- Main difficulties lies in E-step and links to the tractability of the conditional  $p_{\theta}(\mathbf{Z} | \mathbf{X})$ 
  - tractable for mixtures or pPCA
  - intractable for complex model (dependencies, deep-learning part)
- M-step is model dependent, i.e. depends on the choice of  $\mathbf{X} | \mathbf{Z}$ .

**What we did not cover:** practical implementation and caveats

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